

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Summer 2025 ( Supply. )**

**Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 3**

**Subject Code & Name: BTECPC303 - PROGRAMMING, DATA STRUCTURE AND ALGORITHM USING C**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 Which of the following is a non-linear data structure?  
A) Stack                      B) Queue                      C) Linked List                      D) Tree
- What is the process of visiting each element in a data structure exactly once called?  
A) Searching                      B) Sorting                      C) Traversing                      D) Inserting
- What differentiates a circular linked list from a singly linked list?  
A) It has more nodes                      B) Last node points to the first node                      C) It is used only in sorting                      D) First node points to last node
- 4 In a stack implemented using arrays, which operation checks if the stack is full?  
A) Push                      B) StackFull                      C) Top                      D) Pop
- 5 Which of the following expression notations does NOT require parentheses?  
A) Infix                      B) Prefix                      C) Postfix                      D) Both B and C
- 6 Which data structure is most suitable for reversing a list?  
A) Queue                      B) Tree                      C) Stack                      D) Hash Table
- 7 Which of these queue types solves the problem of unused space in a linear queue?  
A) Double-ended queue                      B) Priority queue                      C) Circular queue                      D) Stack
- 8 Which traversal method visits the left subtree, root, and then right subtree  
A) Pre-order                      B) In-order                      C) Post-order                      D) Level-order
- 9 What is the maximum number of children a binary tree node can have?  
A) 1                      B) 2                      C) 3                      D) Unlimited
- 10 Which of the following is used to represent graphs?  
A) Adjacency Matrix                      B) Binary Search Tree                      C) Priority Queue                      D) Hash Table
- 11 What is the worst-case time complexity of linear search in an unsorted array of size n?  
A) O(1)                      B) O(log n)                      C) O(n)                      D) O(n<sup>2</sup>)
- 12 Which of the following sorting algorithms has the best average-case time complexity?  
A) Selection Sort                      B) Insertion Sort                      C) Merge Sort                      D) Bubble Sort

Q2. Solve the following.

- A) Distinguish between linear and non-linear data structures and justify your opinion on 6 which type would be more suitable for a programmer.
- B) Define a linked list. Draw the structure for a singly linked list, a two-way linked list, and a 6 singly circular linked list.

Q3. Solve the following.

- A) Define and represent a Stack in a data structure. Explain the operations of a stack with 6 advantages and disadvantages.
- B) Convert each of the following Infix expressions into Prefix and Postfix expressions 6
  - a)  $A+B/C-D * E$
  - b)  $[(A+B)+C/D * E^F/G]$

Q4. Solve Any Two of the following.

- A) Write the C program for the following operations on a simple queue 6
  - a. Inserting an element
  - b. Deleting an element
- B) What is a queue? Explain three types of it with a suitable diagram. 6
- C) Consider the following queue, where the queue is a Circular Queue having 6 memory 6 cells.  
Front = 2, Rear = 4. Queue:  $\_ , A , C , D , \_ , \_$ .  
Describe the queue as following operation take place : 1) F is added 2) Two letters are deleted 3) R is added to the queue 4) S is added to the queue 5) One letter is deleted.

Q5. Solve Any Two of the following.

- A) Write and explain any three important terminologies used in the tree concept and the 6 graph concept in data structures, with examples.
- B) Explain the Binary Search Tree stepwise with the help of the elements given below 6  
47,12,75,88,90,73,57,1,85,50,62
- C) Explain any one type of graph traversal using an example. 6

Q6. Solve Any Two of the following.

- A) Write an algorithm for 6
  - a. Linear search
  - b. Binary search
- B) Write an algorithm and draw a flow diagram for the Bubble sort technique. 6
- C) Compare the mentioned asymptotic analysis Big-O, Big-Theta, and Big-Omega in 6 evaluating algorithm performance.

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